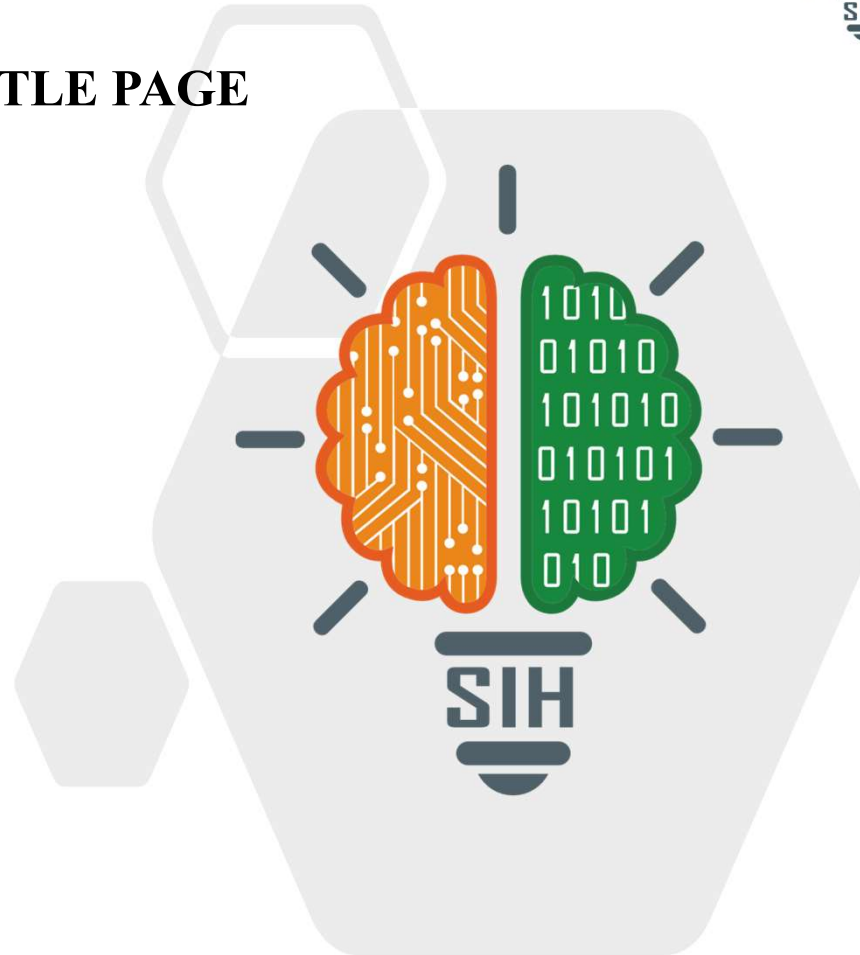


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH-26138
- **Problem Statement Title-**
Dynamic Maritime Decarbonization,
Voyage Speed and Dual-Fuel Optimization
Engine
- **Theme** - Clean and Green Technology
- **PS Category-** Software
- **Team ID-** 145834
- **Team Name** – TeamBuilders



Problem Statement:

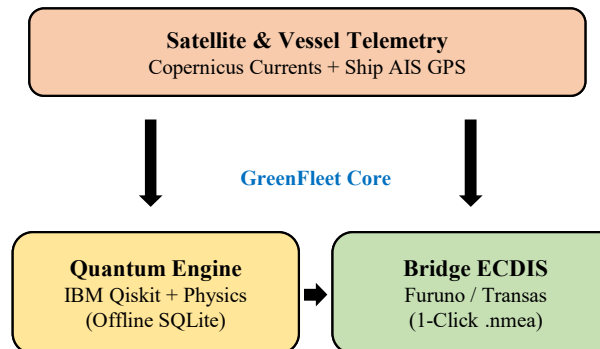
- Fuel takes up over 50% of the entire cost of running a cargo ship.
- Ships sail along fixed paths at constant engine speed, fighting against rough waves and water currents.
- High carbon smoke leads to heavy government fines under new IMO environmental rules.

Our Idea (GreenFleet Quantum):

- A simple software that uses live satellite maps to find the easiest sea path with minimum water resistance.
- Suggests the best engine speed for every part of the trip so the ship burns less fuel.
- Works directly on existing ship computers with zero new hardware and 1-click route loading.
- Cuts fuel burn by **16.8%** and saves over **\$230,000** on a single trip.

Proposed Solution:

- **Autonomous Routing:** Dynamic real-time speed and heading optimization across oceanic corridors using live Copernicus metocean vectors.
- **Naval Physics Twin:** Holtrop-Mennen hull drag, ISO 15016 wave resistance, and non-linear engine SFOC power curves.
- **156-Qubit Hybrid HQOA:** QPSO quantum tunneling + QGA rotation gates evaluating 10^{14} speed
- **Zero-Hardware ECDIS:** NMEA 0183 (\$ECWPL) route export into Furuno, Transas, and Wärtsilä bridge consoles.

Innovation & Uniqueness:

- **Dynamic routes:** The route can change according to real-time ocean and weather conditions.
- **Quantum-powered optimization:** Uses quantum computing to explore a large number of possible routes efficiently.
- **Complete solution:** Combines route planning, fuel reduction, CO₂ reduction, and regulatory compliance in one system.
- **Green fuels :** Ammonia, methane , LPG, etc

TECHSTACK:**Frontend and visualisation:**

- React 19, Typescript, Vite, Leaflet Nautical GIS, Lucide UI

Backend and Physics Engine:

- FastAPI (Python 3.12), Numpy, SciPy, Uvicorn, Pydantic v2

QUANTUM & OPTIMIZATION:

- IBM Qiskit 1.3, QPSO Quantum Tunneling, QGA, PyTorch FNO

METOCEAN AND AIS FEED

- Copernicus CMEMS (1/12), Open-Meteo Maine, AISStream

DATABASES AND SECURITY

- SHA-256 Crypto, SQLite WAL, Other

Tech Stack Symbols:

Python



FastAPI



React 19



TypeScript



IBM Qiskit



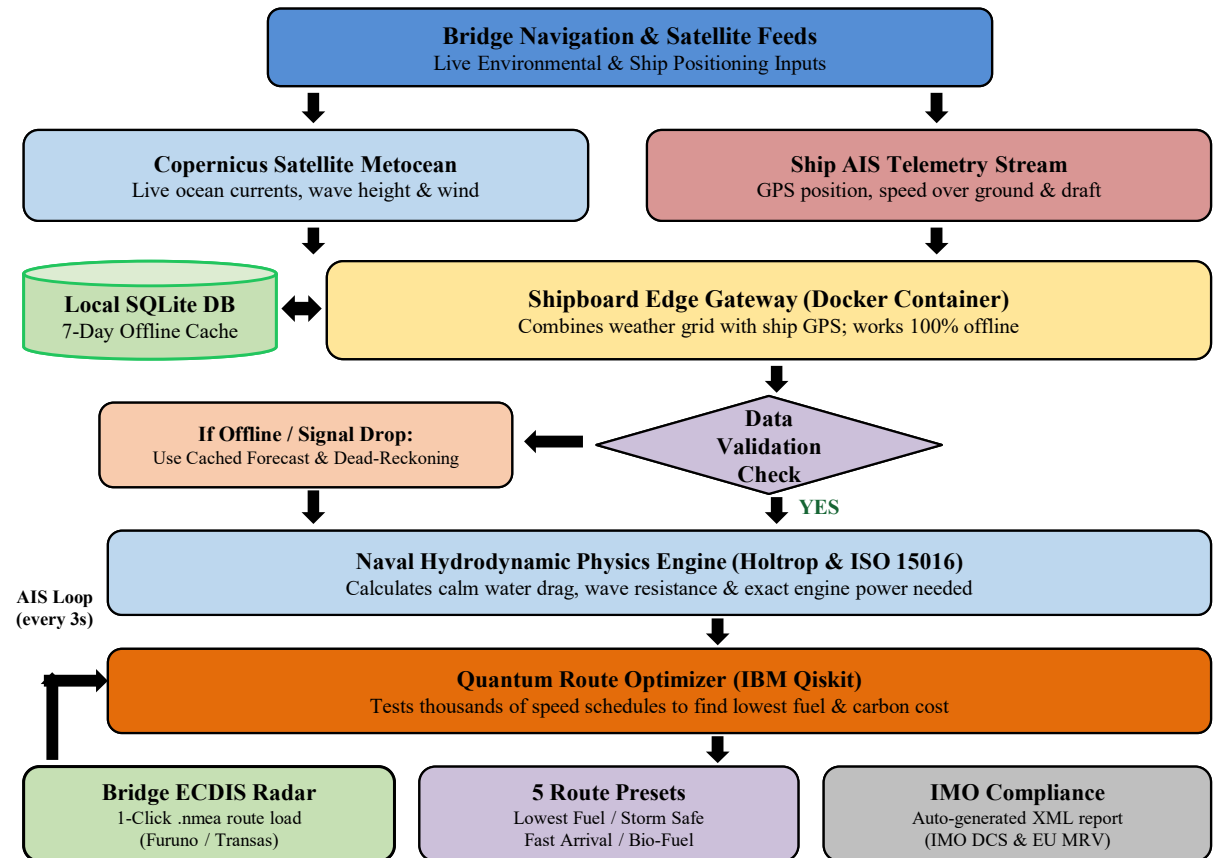
SQLite WAL



Docker



Leaflet GIS

Git Repo:
<https://github.com/thrinadh2005/SIH>
Flowchart (Methodology & Implementation Process):

Feasibility Analysis:

Technical Feasibility:

- Constructed on standard open-source libraries (Python 3.12, FastAPI, React 19, IBM Qiskit), preventing vendor lock-in.
- Hydrodynamic math embeds Holtrop-Mennen calm water regression alongside ISO 15016 wave drag coefficients.
- Direct pipeline ingests free Copernicus Marine (CMEMS) satellite data, eliminating external commercial weather API fees.

[TRL-7 Validated System]

Operational Feasibility:

- Operates as pure containerized software on bridge workstations; requires zero drydocking, hull cuts, or sensor installs.
- Exports standard NMEA 0183 route plans (.nmea) that deck officers load directly onto Furuno, Transas, and Wärtsilä ECDIS.
- Master and watchkeeping officers retain complete navigational discretion, preserving standard bridge operating procedures.

[Zero Capex • Plug-and-Play]

Economic Viability:

- Delivers roughly \$237,800 in net fuel savings on a single Pacific transit for a typical 15,000 TEU container carrier.
- Achieves positive operational cash return on the maiden crossing with zero upfront capital expenditure.
- Lightweight Docker image deploys via vessel IT in under 15 minutes, avoiding shipyard technician callout expenses.

[Breakeven: 1st Voyage]

Potential Challenges & Risks:

- Mid-ocean satellite communication blackouts disrupt links to shorebased cloud servers for hours at a time.
- Sudden gale squalls and high ocean swells drastically increase hull resistance, forcing main engines to burn excess fuel.
- Software or communication failures could stop the system from providing route updates.

Engineering Mitigations & Safeguards:

- A working version can simulate a full journey from Singapore to Rotterdam using real ocean current data and the Holtrop method to calculate ship resistance.
- Helps save fuel and reduce operating costs.
- Can work with existing ship and weather data

Direct Impact on Stakeholders

– Fleet Owners

Each voyage cuts fuel use by 16.8%, saving about \$238,000, with no extra investment required.

– Bridge Officers

Clear RPM guidance for every leg of the trip, plus one-click route loading into navigation systems.

– Bunker Managers

Real-time fuel price tracking at 31 ports helps reduce costs by up to \$40 per ton.

– Regulators & Ports

Automatic, tamper-proof digital logs meet IMO and EU compliance requirements instantly.

Strategic and Safety Value

– IMO CII Grade A Guarantee

Ensures compliance, avoiding trading restrictions and carbon tax penalties.

– Dynamic Wave Drag Avoidance

Routes are adjusted to avoid rough seas, protecting cargo and crew.

– Autonomous Edge Navigation

Works fully offline during ocean crossings, keeping operations safe even without satellite connection.

KEY PERFORMANCE METRICS (VERIFIED SAVINGS):

16.8% LESS FUEL

Fuel Burn Reduction

575 metric tons to be saved / voyage

\$237,800+ SAVED

Reduction of Voyage Cost

Direct bunker cost savings

1,180 TONS CO2 CUT

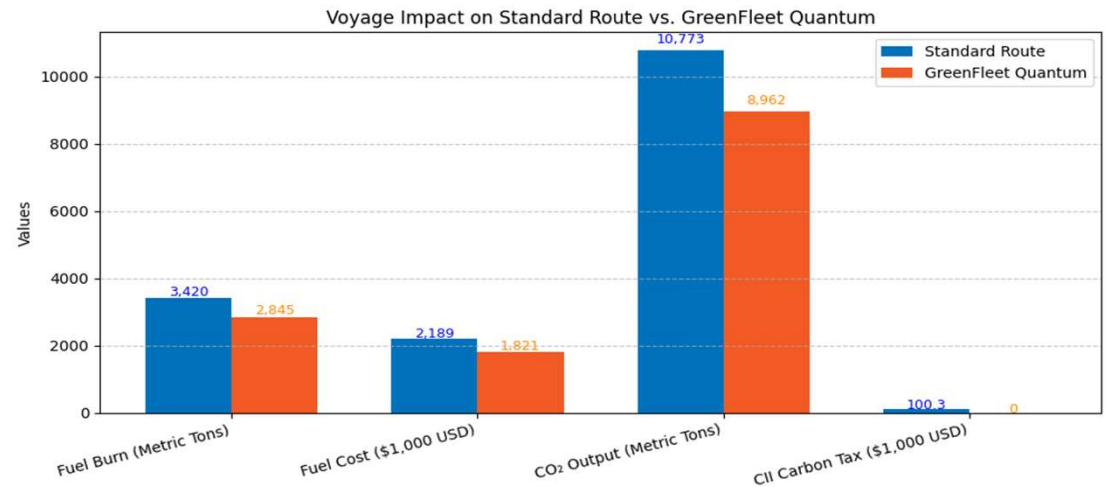
Emissions Abatement

Equivalent to taking ~260 cars off road

IMO GRADE A

Regulatory Compliance

carbon tax penalty exposure to be zeroed



Feature	Current Route Planning	GreenFleet Quantum Platform
Route Planning	Great circle or rhumb lines ignoring surface ocean currents	Dynamic route optimization using Copernicus surface current vectors
Shaft RPM Profile	Constant engine RPM on voyage, regardless of sea state	Variable RPM schedule optimized at each waypoint for minimum fuel consumption
Resistance Modeling	Empirical distance estimation or static weather forecast overlay	Naval hydrodynamic drag models (Holtrop-Mennen & ISO 15016 standards)
Offshore Autonomy	Cloud-reliant systems disrupted during satellite blackouts	Local edge processing with onboard SQLite cache for 100% offline autonomy
Installation & Capex	Requires proprietary sensor retrofit, dry-docking and high capital investment	Software only deployment using Docker container with zero capital expenditure

Details & Links of Reference and Research Work:

[IMO Resolution MEPC.352\(78\)](#) — CII Guidelines & Maritime Decarbonization Targets

[Copernicus Marine Service \(EU\)](#) — Global Ocean Physics Analysis & Forecast (\$1/12^{\circ}\$)

[Holtrop & Mennen \(1982\)](#) — Ship Resistance & Power Prediction Formulation

[ISO 15016:2015](#) — Speed & Power Performance Sea Trial Assessment

[IBM Qiskit Optimization](#) — Transit Scheduling & Combinatorial Quadratic Programming